36 Superplasticity

Superplasticity is the ability of polycrystalline materials to sustain large plastic deformation in tension without rupture, and is phenomenologically delineated by tensile fracture elongations (e_f) in excess of ~300%. Originally discovered in 1912 by Bengough[1], superplasticity is now employed industrially in metal shape forming operations around the world, with primary applications in the aerospace field (*see*, e.g. Ref. [2]).

Superplasticity is a viscous deformation process, and is frequently characterised by a simple flow law relating uniaxial stress σ and strain rate $\dot{\varepsilon}$:

$$\sigma = K \cdot \dot{\varepsilon}^m \tag{1}$$

where K is a temperature- and microstructure-dependent parameter, and m is referred to as the strain rate sensitivity; its reciprocal value, the stress-sensitivity, n, is also commonly reported. Whereas values of m are usually less than 0.25 for typical creep mechanisms in polycrystals, superplasticity is distinguished by values of $m \geq 0.3$. Higher values of m directly promote tensile ductility, by increasing the stability of necks, and by suppressing the tendency for grain-boundary cavitation. In the Newtonian limit of $m = 1$, there is no theoretical limitation to the achievable ductility, although inhomogeneities in temperature or microstructure can substantially impact superplastic extension.

There are two main classes of superplasticity, broadly referred to as *microstructural superplasticity* and *internal-stress superplasticity*, distinguished by deformation mechanisms at the microstructural level.

Microstructural superplasticity is by far the more studied mode of superplastic deformation, and relies on the increase in strain-rate sensitivity, m, observed in fine-grained materials (grain size, d, less than about 10 μm in most metals, and less than about 1 μm in ceramics). At these grain sizes, bulk deformation mechanisms occurring within each grain (i.e. dislocation creep) become secondary to the process of *grain boundary sliding*, involving the relative motion and reorientation of grains with respect to one another. Some general features of microstructural superplasticity are listed below; the reader is referred to Ref. [2] for details:

- The power-law form of Eq. (1), with a constant value of m, does not generally hold over a very broad range of stresses and strain rates in superplastic materials. Rather, a characteristic sigmoidal $\sigma - \dot{\varepsilon}$ relationship is observed, with a peak value of $m \geq 0.3$ occurring over a relatively narrow range of strain rate. This transition is associated with changes in the mechanism for accommodation of grain boundary sliding, from diffusional accommodation at low strain rates to dislocation-motion accommodation at high rates. In superplastic forming operations, it is of primary importance to identify this optimum condition for superplastic elongation.
- The parameter K incorporates an Arrhenius temperature dependence and a power-law grain-size dependence. In general, a finer grain size leads to improved superplastic flow, and the optimum temperature is high enough to promote rapid deformation, but not so high as to encourage rapid grain growth.
- In two-phase superplastic alloys, the optimum superplastic conditions are usually associated with equal proportions of the two phases.
- The optimum strain rate for superplasticity commonly falls in the range 10^{-5}–$10^{-3} s^{-1}$. Recent research has focused on the possibility of high strain rate superplasticity, most commonly observed in aluminium-matrix composite and mechanically alloyed materials at very high temperatures. In such materials, the optimum strain rate may be as high as 10 s^{-1}.
- The fine grain structure required for microstructural superplasticity is usually attained by a schedule of thermomechanical processing, such as rolling or forging, although many more exotic methods such as severe plastic deformation have been developed to refine metallic microstructures.

Table 36.1 summarises 184 investigations of superplasticity in metals and alloys, with an emphasis on the more technologically-relevant aluminium, copper, iron and titanium systems which form the basis of industrial superplastic forming operations. Tabulated are the range of experimental conditions (temperatures, strain rates, and grain sizes), as well as the strain rate sensitivity m, the tensile elongation to failure, e_f, and the optimum conditions for superplasticity. Experimental research in superplasticity has recently focused on emerging structural materials, such as intermetallics, metal-matrix composites, and ceramics. Tables 36.2 and 36.3 summarise recent work on microstructural superplasticity in intermetallics and metal-matrix composites, respectively.

Internal-Stress Superplasticity does not require a fine microstructure, but rather depends upon a renewable source of internal stresses within the material. Internal stresses are generated by, e.g. thermal expansion mismatch between phases or anisotropic grains, or by a polymorphic phase transformation. When an external stress is applied concurrently, the internal mismatch strains are biased to develop preferentially in the direction of the applied external stress, leading to macroscopic plastic deformation. In general, both an internal and a superimposed external stress are required for this deformation mechanism to be active, and the internal stresses must be renewed many times to accumulate large deformations. Thermal cycling is a common means of renewing internal stresses, especially in systems with a thermal expansion mismatch or a thermally-driven phase transformation. Upon each thermal cycle, an increment of plastic strain is developed, and many cycles are required to achieve superplastic strains.

Unlike microstructural superplasticity, which can exhibit a range of strain-rate sensitivities, internal-stress plasticity occurs under ideal Newtonian conditions, with $m = 1$. Unfortunately, this mechanism operates only at low strain rates, in the range 10^{-7}–10^{-4} s^{-1}, but because it does not require a fine microstructure it can be applied to hard-to-form engineering materials such as metal-matrix composites or alloys with coarse microstructures. Internal stress superplasticity has primarily been studied during thermal cycling for the case of thermal expansion-mismatch in aluminium-matrix composites, as well as during allotropic phase transformation (transformation superplasticity) in iron and titanium alloys. Table 36.4 summarises research on internal stress superplasticity.

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	m	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Binary Aluminium Alloys							
Al–6Cu	—	375–455	$7{\cdot}10^{-5}$–$3{\cdot}10^{-2}$	0.21	137	—/425	3
Al–33Cu	1.5–18	380–470	10^{-7}–10^{-2}	0.5	1 300	10^{-7}/380 d<13 μm	4
Al–33Cu	3.4–8.7	450	$7{\cdot}10^{-7}$–$3{\cdot}10^{-2}$	0.7	1 475	$1.3{\cdot}10^{-5}$/—	5
Al–33.23Cu	2–9	360–520	$2{\cdot}10^{-5}$–$3{\cdot}10^{-2}$	0.75	>500	$8{\cdot}10^{-4}$/520	6
Al–2.8Mg	30	200–500	10^{-4}–10^{-1}	0.33	325	10^{-4}/400	7
Al–4Mg	0.5–2	400–538	$8{\cdot}10^{-5}$–$4{\cdot}10^{-2}$	0.3	210	$2{\cdot}10^{-2}$/538	8
Al–5.5Mg	250	200–500	10^{-4}–10^{-1}	0.33	284	$2{\cdot}10^{-2}$/500	7
Al–10Mg	1.6	300	$2{\cdot}10^{-4}$–$8{\cdot}10^{-2}$	—	360	10^{-3}/—	9
Al–0.54Sc	0.5–2	400–538	$8{\cdot}10^{-5}$–$4{\cdot}10^{-2}$	0.25	157	$2{\cdot}10^{-2}$/538	8
Al–13Si	18	490–558	$5{\cdot}10^{-6}$–$2{\cdot}10^{-1}$	0.55	300	$2{\cdot}10^{-1}$/558	10
Al–4Ti	2.6	425–605	[b] 10^{-4}–$3{\cdot}10^{0}$	0.45	—	—	11
Al–10Ti	0.9	600–650	$2{\cdot}10^{-4}$–10^{0}	0.5	493	10^{-1}/650	12
Al–0.5Zr	—	375–455	$7{\cdot}10^{-5}$–$3{\cdot}10^{-2}$	0.36	104	—/423	3
Multicomponent Aluminium Alloys							
Al–4.87Ca–4.64Zn–0.13Fe	1	275–550	10^{-3}–10^{-2}	0.35	305	10^{-3}/550	13
Al–5Ca–4.8Zn	1–5	450–550	$2{\cdot}10^{-3}$–$2{\cdot}10^{-1}$	0.4	900	—	14
Al–5Ca–5Zn	1.2	550–600	10^{-3}–10^{-1}	0.63	900	10^{-1}/600	15
Al–2Cu–0.5Zr	—	375–455	$7{\cdot}10^{-5}$–$3{\cdot}10^{-2}$	0.41	304	—/420	3
Al–2.3Cu–1.7Mg–1.2Ni–1Fe–0.19Si	3	470–550	10^{-3}	1	240	$3{\cdot}10^{-1}$/530	16
Al–3Cu–2Li–1Mg–0.15Zr	2–4	400–500	10^{-4}–$3{\cdot}10^{-2}$	0.4	810	10^{-3}/500	17
	1–7	460–540	10^{-3}–10^{-1}	0.57	1 200	$2{\cdot}10^{-2}$/530	18

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Al–3Cu–2Li–1Mg–0.2Zr	2–4	400–500	$3\cdot10^{-5}$–$3\cdot10^{-2}$	0.4	660	10^{-3}/500	17, 19
Al–4Cu–2Li–0.5Zr	2.3	400–500	$3\cdot10^{-4}$–$3\cdot10^{-2}$	0.4	>1 000	$5\cdot10^{-3}$/450	20
Al–2.4Li–2.6Cu–0.2Zr	0.8–4	450–510	10^{-4}–$2\cdot10^{-2}$	0.6	758	10^{-3}/500	21
Al–4Cu–3Li–0.5Zr	2–8	450	10^{-4}–$3\cdot10^{-2}$	0.45	850	$7\cdot10^{-2}$/—	22
Al–4Cu–1.5Mg–1.1C–0.8O	—	400–500	$8\cdot10^{-4}$–10^{0}	0.3	300	$7\cdot10^{-1}$/425	23
	—	300–550	10^{-3}–$2\cdot10^{2}$	0.5	1 250	$5\cdot10^{1}$/550	24
Al–4Cu–0.5Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.43	418	—/420	3
	0.3–8	220–500	$3\cdot10$–$4\cdot10^{0}$	0.5	800	—/500, d = 8 μm	25
Al–6Cu–0.2Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.45	191	—/400	3
Al–6Cu–0.26Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.41	233	—/425	3
Al–6Cu–0.33Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.44	554	—/450	3
Al–6Cu–0.41Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.42	612	—/450	3
Al–6Cu–0.5Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.39	612	—/455	3
	—	375–500	$2\cdot10^{-4}$–$2\cdot10^{-2}$	0.4	600	$2\cdot10^{-2}$/600	26
	1.5–10	300–505	$1\cdot10^{-4}$–$2\cdot10^{-3}$	—	508	—	27
Al–12Cu–0.5Zr	—	375–455	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.35	390	—/455	3
Al–25Cu–11Mg	5–6	420–480	$4\cdot10^{-6}$–$3\cdot10^{-3}$	0.7	>600	[a] >10^{-3}/450–80	28
Al–33Cu–7Mg	5–6	420–480	10^{-6}–10^{-2}	0.72	>600	[a] >10^{-3}/450–80	28
Al–3Li–0.5Zr	—	450	10^{-4}–$3\cdot10^{-2}$	0.45	1 035	$3\cdot10^{-2}$/—	22
Al–3.2Li–1Mg–0.3Cu–0.18Zr	<4	490–580	10^{-3}–10^{0}	0.5	250	$1\cdot10^{-1}$/570	29
Al–2.7Mg–2.04Li–0.5Cu–0.13Zr	0.5	350–500	$5\cdot10^{-4}$–10^{-1}	0.5	670	$8\cdot10^{-4}$/450	30
Al–4Mg–1.5Li–1.2C–0.4O	0.4	425–575	10^{-3}–$2\cdot10^{2}$	0.5	180	$2\cdot10^{1}$/575	31
Al–4Mg–0.56Sc	0.5–2	400–538	$8\cdot10^{-5}$–$4\cdot10^{-2}$	0.4	>1 020	$2\cdot10^{-2}$/538	8
Al–4Mg–1.5O–0.8C	0.3	25–325	$5\cdot10^{-4}$–$5\cdot10^{-2}$	—	130	$5\cdot10^{-2}$/300	32
Al–4.3Mg–0.2Zr	10	475	$4\cdot10^{-5}$–$4\cdot10^{0}$	0.15	130	$4\cdot10^{0}$/—	33
Al–4.3Mg–0.3Zr	1	475	10^{-4}–$4\cdot10^{0}$	0.3	220	10^{-2}/—	33
Al–4.3Mg–0.8Zr	1	475	10^{-4}–$4\cdot10^{0}$	0.3	450	10^{-2}–10^{-1}/—	33
Al–5Mg–1.2Cr	3	372–550	10^{-4}–10^{0}	0.5	1 000	$2\cdot10^{-2}$/550	34
Al–5Mg–2Li–0.1Zr	0.1	250–350	10^{-1}	—	350	—/250	35
	0.3–10	180–400	$5\cdot10^{-4}$	0.42	320	—/400, d = 10 μm	25
Al–5Mg–1.5Mn	5	550	10^{-4}–$2\cdot10^{0}$	0.45	500	10^{-2}/—	33
Al–5Mg–2.2Mn	3	550	10^{-4}–$2\cdot10^{0}$	0.45	400	$2\cdot10^{-3}$/—	33
Al–5Mg–0.2Zr	8	500	$2\cdot10^{-5}$–$4\cdot10^{0}$	0.15	150	$3\cdot10^{-4}$/—	33
Al–5Mg–0.5Zr	1	500	$1\cdot10^{-4}$–$4\cdot10^{0}$	0.3	400	10^{-2}–10^{-1}/—	33
Al–5Mg–0.8Zr	1	500	$1\cdot10^{-4}$–10^{0}	0.3	575	10^{-1}/—	33
Al–5.5Mg–2.2Li–0.12Zr	1.2	350	10^{-2}–10^{0}		>1 180	10^{-2}/—	36
Al–5.76Mg–0.32Sc–0.3Mn–0.1Fe–0.2Si–0.1Zn	3	450–560	10^{-4}–10^{0}	0.6	1 130	$1\cdot10^{-2}$/475	37
Al–6Mg–0.54Sc	0.5–2	316–538	$8\cdot10^{-5}$–$4\cdot10^{-2}$	—	>1 020	$2\cdot10^{-2}$/538	8
Al–10Mg–0.5Mn	0.5	200–425	$7\cdot10^{-5}$–$1\cdot10^{-1}$	0.45	700	$6\cdot10^{-3}$/300	38
Al–9.89Mg–0.09Zr	2–5	300	$7\cdot10^{-5}$–$2\cdot10^{-2}$	0.45	600	$2\cdot10^{-3}$/—	39
Al–10Mg–0.1Zr	2	150–400	$7\cdot10^{-5}$–$7\cdot10^{-2}$	0.5	1 100	10^{-3}/300	40
	1.5	300	$2\cdot10^{-4}$–$8\cdot10^{-2}$	—	420	10^{-3}/—	9
	2	200–425	$7\cdot10^{-5}$–$1\cdot10^{-1}$	0.5	1 100	$7\cdot10^{-3}$/300	38
Al–10Mg–0.4Zr	0.9	300	$2\cdot10^{-4}$–$8\cdot10^{-2}$	—	640	10^{-3}/—	9
Al–10Mg–0.6Zr	0.7	300	$2\cdot10^{-4}$–$8\cdot10^{-2}$	—	600	10^{-2}/—	9
Al–10.2Mg–0.52Mn	5	25–425	10^{-4}–10^{-1}	0.45	580	$5\cdot10^{-3}$/300	41
Al–9Zn–1Mg	9.8	520	$5\cdot10^{-4}$–$5\cdot10^{-1}$	0.65	>200	$5\cdot10^{-3}$/—	42

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Engineering Aluminium Alloys							
Al-2004 (Supral 100): Al–6Cu–0.4Zr–0.25Mg–0.1Fe–0.05Zn–<0.1Mn							
	0.5	300–350	10^{-2}–10^{-1}	—	970	10^{-2}/300	36
Supral 220: Al–6Cu–0.4Zr–<0.2 Si, Ge, Zn, Fe							
	4	460	$5{\cdot}10^{-5}$–$5{\cdot}10^{-2}$	—	1 100	—	43
	—	480	10^{-3}–10^{-1}	0.451	—	—	44
Al-2024: Al–4.4Cu–1.5Mg–0.6Mn (nominal)							
	5–8	430–485	$2{\cdot}10^{-4}$–$2{\cdot}10^{-2}$	0.6	485	$4{\cdot}10^{-4}$/480	45
+0.7Mg	6.6	475–490	10^{-5}–$4{\cdot}10^{-2}$	0.6	400	$5{\cdot}10^{-4}$/—	46
Al-2090: Al–2.8Cu–2.2Li–0.25Mg–0.15Zr (nominal)							
	10	518	$4{\cdot}10^{-4}$–10^{-3}	—	1 373	$4{\cdot}10^{-4}$/—	47
	—	485	$5{\cdot}10^{-5}$–10^{-2}	0.42	850	$3{\cdot}10^{-4}$/—	48
	10–15	493–527	$2{\cdot}10^{-4}$–10^{-2}	0.5	1 690	$2{\cdot}10^{-4}$/510	49
	20	510–530	$2{\cdot}10^{-5}$–10^{-2}	0.6	900	$2{\cdot}10^{-3}$/510	50
Al-2195 (Weldalite): Al–4.8Cu–1.3Li–0.4Mg–0.4Ag–0.14Zr (nominal)							
	<20	463–496	$5{\cdot}10^{-5}$–10^{-3}	0.64	700	$5{\cdot}10^{-4}$/485	51
Al-5056: Al–5Mg–0.12Mn–0.12Cr (nominal)							
	0.3	200–275	10^{-5}–10^{-4}	0.3	185	10^{-5}/275	52
Al-5083: Al–4.4Mg–0.7Mn–0.12Cr (nominal)							
	6–14	500–550	$5{\cdot}10^{-4}$–10^{-3}	0.65	450	$5{\cdot}10^{-4}$/550	53
	6.7	450–550	10^{-4}–10^{-1}	0.58	425	$4{\cdot}10^{-4}$/550	54
	6–10	510	10^{-5}–10^{-2}	0.48	320	$4{\cdot}10^{-4}$/—	55
	8.5–9.5	510–550	$2{\cdot}10^{-4}$–10^{-3}	0.57	450	$2{\cdot}10^{-4}$/510	56
	7–14	450–570	$3{\cdot}10^{-5}$–$3{\cdot}10^{-3}$	0.75	700	$2{\cdot}10^{-4}$/525	57
	9–39	500–565	10^{-5}–10^{-2}	0.65	630	10^{-2}/550	58
+0.19Zr	7.5	450–550	10^{-4}–10^{-1}	0.55	350	$4{\cdot}10^{-4}$/550	54
+1.60Mn	5.9	450–550	10^{-4}–10^{-1}	0.6	425	$2{\cdot}10^{-3}$/550	54
+1.57Mn, 0.2Zr	4.8	450–550	10^{-4}–10^{-1}	0.65	425	$2{\cdot}10^{-3}$, 550	54
Al-5454: Al–2.69Mg–0.74Mn–0.12Si (nominal)							
	10–18	500–550	$5{\cdot}10^{-4}$–10^{-3}	0.44	239	$5{\cdot}10^{-4}$, 500	53
Al-5456: Al–5.1Mg–0.8Mn–0.12Cr (nominal)							
	20	25–400	10^{-5}–10^{-2}	0.42	150	$2{\cdot}10^{-5}$, 275	59
Al-5754: Al–3.15Mg–0.25Mn–0.18Fe (nominal)							
	21–30	500	10^{-3}	0.32	140	—	53
Al-6061: Al–1.0Mg–0.6Si–0.28Mn–0.20Cr (nominal)							
	200	500–610	10^{-5}–$2{\cdot}10^{-2}$	0.9	350	10^{-4}, 590	60
	10	475–600	$3{\cdot}10^{-4}$	—	1 250	—/590	60
Al-7075: Al–5.6Zn–2.5Mg–1.6Cu–0.23Cr (nominal)							
	4	440–520	10^{-5}–10^{-2}	0.63	500	10^{-3}–10^{-2}/—	9
	10	480–520	10^{-4}–$3{\cdot}10^{-3}$	0.75	1 200	—	61
	10–17	510	$6{\cdot}10^{-4}$–$9{\cdot}10^{-3}$	—	2 100	$8{\cdot}10^{-4}$/—	62
Al-7091: Al–6.5Cu–2.5Mg–1.5Cu–0.4Co (nominal)							
	3–5	300–500	$8{\cdot}10^{-5}$–$2{\cdot}10^{-2}$	0.38	445	$8{\cdot}10^{-5}$, 300	63
Al-7475: Al–5.7Zn–2.2Mg–1.6Cu–0.22Cr (nominal)							
	2–7	515	10^{-3}–10^{-1}	0.4	1 750	10^{-1}/—	64
	4–7	515	$2{\cdot}10^{-5}$–$2{\cdot}10^{-2}$	0.9	1 020	$8{\cdot}10^{-4}$/—	65
	5.5–13	516	$2{\cdot}10^{-4}$–$2{\cdot}10^{-2}$	0.67	2 000	$2{\cdot}10^{-3}$/— d = 5.5 μm	66

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
	6.8	505	10^{-5}–$6\cdot10^{-3}$	0.5	>400	—	67
	9	437–516	$^{e}10^{-4}$–10^{-2}	0.58	550	10^{-4}/516	68
	10	516	$2\cdot10^{-4}$–$5\cdot10^{-3}$	0.4	500	—	69
	11	400–516	$6\cdot10^{-6}$–10^{-2}	0.8	780	$2\cdot10^{-4}$/516	70
	11	400–516	$6\cdot10^{-5}$–10^{-2}	0.67	780	10^{-4}/516	71
	12.3	371–516	$2\cdot10^{-5}$–10^{-2}	0.8	1 400	$2\cdot10^{-4}$/516	72
	16.3	371–516	$2\cdot10^{-5}$–10^{-2}	0.9	850	$2\cdot10^{-4}$/516	72
	15–20	516	$6\cdot10^{-5}$–$2\cdot10^{-3}$	0.5	500	—	73
	22	371–516	$2\cdot10^{-5}$–10^{-2}	0.9	730	$2\cdot10^{-4}$/516	72
	69.4	371–516	$2\cdot10^{-5}$–10^{-2}	0.75	210	$2\cdot10^{-4}$/516	72
	156.3	371–516	$2\cdot10^{-5}$–10^{-2}	0.5	170	$2\cdot10^{-4}$/516	72
+0.72Zr	1–2.8	520	$5\cdot10^{-3}$–$3\cdot10^{-1}$	0.62	900	$5\cdot10^{-2}$/—	74, 75
+1.2Fe, 0.1Si	—	516	$5\cdot10^{-4}$–$5\cdot10^{-2}$	0.6	—	—	44
Al-8090: Al–2.5Li–1.3Cu–0.8Mg–0.12Zr–0.15Fe–0.04Si (nominal)							
	1	300–525	$8\cdot10^{-5}$–$5\cdot10^{-3}$	0.5	645	$8\cdot10^{-4}$/350	76
	2	450–550	$4\cdot10^{-5}$–$6\cdot10^{-3}$	0.5	760	$6\cdot10^{-4}$/500	77
	3	300–540	$\geq 2\cdot10^{-4}$	0.6	550	$8\cdot10^{-4}$/525	78
	2–4	530	$5\cdot10^{-4}$–$5\cdot10^{-2}$	0.45	—	—	44
	1.6–300	460–540	$3\cdot10^{-5}$–$3\cdot10^{-2}$	—	475	$3\cdot10^{-4}$/540	79
	14	518	$2\cdot10^{-4}$–10^{-3}	—	963	$4\cdot10^{-4}$/—	47
Al–9021: Al–4Cu-1.5Mg–1.1C–0.8O (nominal)							
	0.5	550	10^{-3}–$3\cdot10^{2}$	0.5	1 100	10^{2}/—	80
	0.2–3	424–475	10^{-4}–$3\cdot10^{-2}$	0.25	525	$2\cdot10^{0}$/475	81
Al-9052: Al–4Mg–1.1C–0.8O (nominal)							
	0.5	500–590	10^{-3}–10^{2}	0.5	330	10^{1}/590	82
LFC-X1: Al–(5–7)Mg–(1–2)Zn–(0.5–1)Mn–(0.1–0.2)Cr–(0.3–0.5)Zr							
	<10	510–550	$4\cdot10^{-5}$–$8\cdot10^{-4}$	0.6	800	$3\cdot10^{-4}$/—	83
Formall 570: Al–4.5Mg–3.5Zn–0.35Mn–0.25Cr–0.1Si–0.1Fe							
	7	485–530	$6\cdot10^{-5}$–$2\cdot10^{-2}$	0.6	600	$2\cdot10^{-4}$/515	84
Al-PM64: Al–7.15Zn–2.39Mg–2.02Cu–0.09Fe–0.19Zr–0.12Cr–0.20Co							
	6	500–516	10^{-5}–$2\cdot10^{-2}$	0.85	800	$2\cdot10^{-4}$/—	85
Copper Alloys							
Cu–2.8Al–1.8Si–0.4Co	1.25	550	$5\cdot10^{-4}$–$2\cdot10^{-2}$	0.42	500	$2\cdot10^{-4}$/—	86
	2.8	550	10^{-6}–10^{0}	0.5	2 550	$7\cdot10^{-2}$/—	87
Cu–8.2Al–4.4Fe–5.2Ni–1.5Mn	—	800	$2\cdot10^{-5}$–$6\cdot10^{-1}$	0.52	>1 300	$2\cdot10^{-3}$/—	88
Cu–8.9Al–4.1Fe–3.9Ni–0.9Mn	2	750–850	$2\cdot10^{-5}$–$6\cdot10^{-1}$	0.68	650	10^{-3}/—	88
Cu–9.1Al–5.9Fe–5.1Ni–1.6Mn	—	800	$2\cdot10^{-5}$–$6\cdot10^{-1}$	0.56	>1 300	$8\cdot10^{-3}$/—	88
Cu–10Al–4.5Fe–6.0Ni–1.7Mn	2	800	$2\cdot10^{-5}$–$6\cdot10^{-1}$	0.57	>1 300	$6\cdot10^{-3}$/—	88
Cu–10.3Al–5.2Fe–5.0Ni–1.5Mn	2–4	750–850	$2\cdot10^{-5}$–$6\cdot10^{-1}$	0.68	>5 500	$6\cdot10^{-3}$/750	88, 89
Cu–13.5Al–3Ni–(X)Cr (X = 0–1)	3–6	680	$2\cdot10^{-3}$	—	620	X = 0.5	90
Cu–14Al–3Ni–(X)Cr (X = 0–1)	3–6	680	$2\cdot10^{-3}$	—	1 000	X = 0.5	90
Cu–1.9Be–0.2Ni	—	550	$2\cdot10^{-3}$	—	1 300	—	91
Cu–21.25Zn–6.58Al–3.13Fe–2.32Mn	70	550–750	10^{-3}–$4\cdot10^{-2}$	—	377	$3\cdot10^{-2}$/700	92

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Cu–28.1Zn–15.0Ni–13.3Mn	1–3	570	10^{-5}–$2\cdot10^{-3}$	0.5	620	$2\cdot10^{-4}$/—	93, 94
Cu–38.4Zn–15.7Ni–0.13Mn	1–3	462–652	10^{-5}–$2\cdot10^{-3}$	0.5	750	$6\cdot10^{-4}$/580	93, 94
Cu–38.5Zn–3Fe	10–80	500–800	$2\cdot10^{-5}$–10^{-2}	0.5	350	$2\cdot10^{-4}$/700	95
Cu–40Zn	3	560–640	$3\cdot10^{-5}$–$6\cdot10^{-3}$	0.63	410	—	96
Cu–40Zn–1.5Pb	2 000–3 000	670–720	$4\cdot10^{-4}$–$2\cdot10^{-3}$	0.56	192	$6\cdot10^{-4}$/700	97
Iron Alloys							
Fe–0.03C–4Ni–3Mo–1.6Ti	6	900–960	$4\cdot10^{-5}$–$4\cdot10^{-3}$	0.53	>500	$8\cdot10^{-4}$/800	98
Fe–0.03C–4Ni–3Mo–1.6Ti + 600 ppm B	6	900–960	$4\cdot10^{-5}$–$4\cdot10^{-3}$	0.53	634	$2\cdot10^{-4}$/900	98
Fe–(0.13–0.34)C	2–12	700–950	$3\cdot10^{-5}$–$7\cdot10^{-3}$	0.58	376	$3\cdot10^{-4}$/900	99
Fe–0.42C–1.9Mn	1.4–2.6	718–782	$2\cdot10^{-6}$–$2\cdot10^{-2}$	0.65	460	$3\cdot10^{-4}$/727	100
Fe–0.88C–1.15Mn–0.22V–0.5W–0.48Cr–0.31Si	<3	650	$4\cdot10^{-5}$–10^{2}	0.5	1 200	$2\cdot10^{-4}$/—	101
Fe–0.91C–0.45Mn–0.12Si	1–5	716–927	10^{-3}–10^{2}	0.42	133	$4\cdot10^{-5}$/716	102
Fe–1.11C–0.48Mn–1.45Cr–9.30Al	3–4	700–900	$7\cdot10^{-6}$–10^{-2}	0.74	—	—	103
Fe–1.17C–5.12Al	1–2	750–850	10^{-4}–$8\cdot10^{-2}$	0.64	490	$2\cdot10^{-3}$/850	104
Fe–1.25C–0.65Mn–0.1Si	1.5–3	605–800	10^{-4}–$2\cdot10^{-2}$	0.5	700	10^{-4}/630	105
Fe–1.25C–10Al–1.5Cr–0.54Mn	10	750–920	10^{-5}–$2\cdot10^{-2}$	0.74	>700	$5\cdot10^{0}$/920	106
Fe–1.32C–0.6Mn–1.68Cr–1.9Al	1–2	700–780	$7\cdot10^{-6}$–10^{-2}	0.43	—	—	103
Fe–1.34C–3.05Al	1–2	750–850	10^{-4}–$8\cdot10^{-2}$	0.5	583	$8\cdot10^{-3}$/750	104
Fe–1.51C–3.38Al	1–2	750–850	10^{-4}–$8\cdot10^{-2}$	0.49	783	$8\cdot10^{-3}$/750	104
Fe–1.55C–5.22Al	1–2	750–850	10^{-4}–$8\cdot10^{-2}$	0.63	450	$2\cdot10^{-3}$/850	104
Fe–1.6C–0.73Mn–0.28Si	1.5–3	600–800	$7\cdot10^{-5}$–$3\cdot10^{-2}$	0.5	760	10^{-4}/630	105
Fe–1.92C–0.82Mn–0.3Si	1.5–3	600–800	$2\cdot10^{-4}$–$2\cdot10^{-3}$	0.5	380	$2\cdot10^{-4}$/650	105
Fe–2.4C	2	630–725	$2\cdot10^{-5}$–$5\cdot10^{-3}$	0.5	500	$2\cdot10^{-4}$/700	107
Fe–3C	2	650–700	$2\cdot10^{-5}$–$5\cdot10^{-4}$	0.5	940	$2\cdot10^{-4}$/700	107
Fe–3C–1.5Cr	1	630–725	$2\cdot10^{-5}$–$5\cdot10^{-3}$	0.5	1 410	$2\cdot10^{-4}$/700	107
Fe–3.26C–4.1N–2.1Cr–0.38Si–0.45Mn	1–2	650–750	$7\cdot10^{-5}$–$2\cdot10^{-3}$	0.46	210	$2\cdot10^{-4}$/675	108
Fe–4.3C–0.18Si	1–2	650–750	$9\cdot10^{-5}$–$3\cdot10^{-3}$	0.41	150	$2\cdot10^{-4}$/675	108
Stainless Steels							
Fe–20Cr–10Ni–0.7Ni	—	700–1 000	$3\cdot10^{-4}$–$2\cdot10^{-2}$	0.65	527	—/800	109
Fe–21.3Cr–4.2Mn–2.1Ni–0.1Si–0.74Cu	10–20	900–1 050	10^{-4}–$6\cdot10^{-2}$	0.56	—	—	110
Fe–23Cr–4Ni–0.1N	7.5	880–1 000	$5\cdot10^{-6}$–$3\cdot10^{-2}$	0.8	—	—	111
Fe–24.66Cr–6.82Ni–2.79Mo–0.85Mn–0.48Si–0.28W–0.14N	<20	900–1 050	$4\cdot10^{-4}$–$4\cdot10^{-2}$	0.45	2 000	$4\cdot10^{-3}$/900	112
Fe–25Cr–7Ni–3Mo	5	900–1 050	10^{-4}–$6\cdot10^{-2}$	0.89	—	—	110
	1–2	1 000	10^{-4}–$2\cdot10^{0}$	0.66	>1 600	10^{-2}/—	113

(continued)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Fe–25Cr–7Ni–3Mo–0.14N (SuperDux 64)	2–3	1 050–1 350	10^{-3}–10^{0}	0.6	1 000	$2{\cdot}10^{-2}$/1350	114
Fe–26Cr–6.5Ni–0.4Ti–0.5Mn–0.5Si	3–4	700–1 020	$2{\cdot}10^{-6}$–$2{\cdot}10^{-2}$	0.54	1 050	$3{\cdot}10^{-3}$/960	115
Magnesium Alloys							
Mg–2.83Al–0.09Zr (AK20)	73.1	300–500	$8{\cdot}10^{-5}$–$8{\cdot}10^{-2}$	—	330	$8{\cdot}10^{-4}$/450	116
Mg–5.14Al–0.18Zr (AK50)	43.9	300–500	$8{\cdot}10^{-5}$–$8{\cdot}10^{-2}$	—	350	$8{\cdot}10^{-4}$/450	116
Mg–6.19Al	66.8	300–500	$8{\cdot}10^{-5}$–$8{\cdot}10^{-2}$	—	260	$8{\cdot}10^{-4}$/450	116
Mg–8.3Al–8.1Ga	0.2	250–350	10^{-4}–10^{0}	0.5	1 080	10^{-2}/250	117
Mg–9Al–0.7Zr–0.15Mn (AK91)	1–20	250–300	$4{\cdot}10^{-4}$–10^{-2}	0.62	1 480	$4{\cdot}10^{-1}$/300, d = 1 μm	118
	1	175–200	$2{\cdot}10^{-5}$–10^{-3}	0.33	661	$6{\cdot}10^{-5}$/200	119
	0.5–3	150–250	$2{\cdot}10^{-5}$–$3{\cdot}10^{-4}$	0.5	950	$7{\cdot}10^{-5}$/200	120
	<5	270–400	$2{\cdot}10^{-5}$–10^{0}	0.5	425	$3{\cdot}10^{-4}$/250	121
Mg–11.28Al–0.42Zr	19.7	300–500	$8{\cdot}10^{-5}$–$8{\cdot}10^{-2}$	—	110	$8{\cdot}10^{-4}$/450	116
Mg–33.6Al	1–2	250–400	$3{\cdot}10^{-7}$–$3{\cdot}10^{-2}$	0.85	2 100	$3{\cdot}10^{-2}$/400	122
Mg–6.5Li	10.5	125–300	10^{-5}–$8{\cdot}10^{-3}$	0.33	—	—	123
Mg–9Li	2.7	25–175	$7{\cdot}10^{-6}$–$5{\cdot}10^{-3}$	0.5	450	$6{\cdot}10^{-4}$/100	124
	6–34	150–250	$4{\cdot}10^{-5}$–$3{\cdot}10^{-2}$	0.5	460	$3{\cdot}10^{-4}$/180, d = 6 μm	125
Mg–5.8Zn–0.65Zr (ZK60)	0.8–14	150–450	10^{-5}–10^{-1}	0.38	700	10^{-4}/250	126
	<2	250–350	10^{-3}–10	0.5	425	10^{-1}/400	121
Mg–6Zn–0.5Zr (ZK60)	3.4	150–400	10^{-5}–10^{-2}	0.5	540	10^{-2}/325	127
	0.8–10	270	$3{\cdot}10^{-6}$–$6{\cdot}10^{-2}$	0.52	—	[a] 10^{-4}/—, d = 4 μm	128
	6.5	200–250	$3{\cdot}10^{-6}$–$3{\cdot}10^{-5}$	0.5	450	$<1{\cdot}10^{-5}$/—	129
	<7	250–450	$2{\cdot}10^{-4}$–10^{-1}	0.5	725	$4{\cdot}10^{-3}$/300	121
Mg–6Zn–0.65Zr	5–12	250–500	10^{-4}–$4{\cdot}10^{-3}$	0.38	—	—	130
Nickel Alloys							
Ni–12.1Cr–5.96Al–0.78Ti–4.61Mo–0.13Zr	1.3–3	1 050–1 100	[b] 10^{-5}–10^{0}	0.5	—	—	131
Ni–12.38Cr–3.53Mo–18.52Co–4.26Ti–4.90Al	3	1 050–1 100	[b] $8{\cdot}10^{-5}$–$4{\cdot}10^{-3}$	0.6	—	[a] $6{\cdot}10^{-4}$/1075	132, 133
Ni–12.5Cr–18.3Co–4.4Ti–3.2Mo–4.9Al–0.76V	2.1	900–1 040	$5{\cdot}10^{-5}$–$6{\cdot}10^{-1}$	0.66	>850	$<10^{-1}$/—	134
Ni–13.1Cr–10.7Co–4.6Mo–2.8W–3.2Al–3.4Nb–2.6Ti	0.25–5.5	800–1 075	$7{\cdot}10^{-5}$–10^{-2}	0.6	>550	10^{-3}/1075, d = 5.5 μm	135
Ni–15Cr–4.5Al–2.5Ti–2Mo–4W–2Ta/1.1 vol% Y_2O_3 [c]	0.26	900–1 100	$3{\cdot}10^{-8}$–$6{\cdot}10^{0}$	0.47	308	$5{\cdot}10^{-1}$/1 000	136
Ni–20Cr–0.5Ti–0.3Al/0.6 vol% Y_2O_3 [c]	0.7	900–1 100	$2{\cdot}10^{-4}$–$3{\cdot}10^{0}$	0.3	200	10^{-1}/1 000	136
Ni–20Cr–18Fe–5Nb–3Mo–1Co (Inconel 718)	<6	982	10^{-4}	0.75	500	—	137, 138
	1–10	900–980	$2{\cdot}10^{-4}$–$5{\cdot}10^{-3}$	0.77	1 050	10^{-3}/920	139
	5–12	927–982	$4{\cdot}10^{-5}$–10^{-2}	0.9	760	$3{\cdot}10^{-5}$/954	140
Ni–29.97Cr–5.27Al	0.6–1.5	852–1 002	$3{\cdot}10^{-4}$–$2{\cdot}10^{-2}$	0.5	500	$2{\cdot}10^{-3}$/852	141
Ni–39Cr–9Fe–2Ti–1Al	<5	816–982	$2{\cdot}10^{-5}$–$2{\cdot}10^{-1}$	0.5	>880	$2{\cdot}10^{-3}$–$2{\cdot}10^{-2}$/927–982	142

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Tin Alloys							
Sn–1Bi	2–20	25	$2\cdot10^{-8}$–$2\cdot10^{-2}$	0.45	300	$2\cdot10^{-4}$/—	143
Sn–32Pb–5.2Bi	1.5	25	$5\cdot10^{-6}$–10^{-2}	0.48	1 275	10^{-2}/—	144
Sn–35.4Pb–6.9Ag	1.7	25	$5\cdot10^{-6}$–10^{-2}	0.4	350	$3\cdot10^{-5}$/—	144
Sn–35.4Pb–7.4Sb	11	25	$5\cdot10^{-6}$–10^{-2}	0.4	490	10^{-3}/—	144
Sn–36Pb–5.6Ag	2–3	25	$4\cdot10^{-5}$–$4\cdot10^{-3}$	0.4	362	$8\cdot10^{-4}$/—	145
Sn–36.6Pb–4.1Ag	1.9	25	$5\cdot10^{-6}$–10^{-2}	0.42	550	10^{-3}/—	144
Sn–36.6Pb–4.3Cu	2.3	25	$5\cdot10^{-6}$–10^{-2}	0.43	220	10^{-3}–10^{-4}/—	144
Sn–36.6Pb–4.1Sb	1	25	$5\cdot10^{-6}$–10^{-2}	0.41	675	$3\cdot10^{-4}$/—	144
Sn–38.1Pb	2	25	$4\cdot10^{-5}$–$4\cdot10^{-3}$	0.53	1 812	$8\cdot10^{-4}$/—	145
	2.2	25	$5\cdot10^{-6}$–10^{-2}	0.45	—	—	144
	6.9	140	10^{-4}	—	>4 850	—	146
Sn–9.8Zn	1–2	121–177	$3\cdot10^{-6}$–10^{-3}	0.5	570	$2\cdot10^{-4}$/163	147
Titanium Alloys							
Ti–2.5Al–0.5Mn–0.3Zr–<0.3Fe	15–20	800–900	$4\cdot10^{-5}$–$5\cdot10^{-3}$	0.3	340	10^{-3}/850	77
Ti–4Al–4Mo–2Sn–0.5Si	2–5	805–915	10^{-5}–10^{-2}	0.8	1 230	-/915	148
Ti–4Al–0.25O	23.5	800–1 025	$4\cdot10^{-5}$–$4\cdot10^{-4}$	0.6	—	—	149
Ti–4.5Al–3V–2Fe–2Mo	4	700–850	$2\cdot10^{-5}$–$2\cdot10^{-3}$	0.5	300	$2\cdot10^{-5}$–10^{-3}/—	150
Ti–5Al–2.5Sn	18.5	800–1 055	$3\cdot10^{-5}$–10^{-2}	0.72	>400	—	149
Ti–5.8Al–4.02Sn–3.49Zr–0.71Nb–0.52Mo–0.33Si–0.05C	8–20	990	$2\cdot10^{-5}$–$2\cdot10^{-3}$	0.7	>300	10^{-4}/—	151
Ti–6Al–3.2Mo	0.06	25–600	$3\cdot10^{-8}$–$2\cdot10^{-4}$	0.35	600	$5\cdot10^{-4}$/600	152
	0.1	25–600	$3\cdot10^{-8}$–$2\cdot10^{-4}$	0.33	480	$5\cdot10^{-4}$/600	152
	0.4	25–600	$3\cdot10^{-8}$–$2\cdot10^{-4}$	0.3	225	$5\cdot10^{-4}$/600	152
	5	25–600	$3\cdot10^{-8}$–$2\cdot10^{-4}$	0.1	35	$5\cdot10^{-4}$/600	152
Ti–6Al–4V	3	750–925	$1\cdot10^{-5}$–10^{-2}	0.71	490	10^{-4}/750	153
	4.5	760–940	$2\cdot10^{-4}$–$2\cdot10^{-3}$	0.75	1 050	$2\cdot10^{-4}$/880	154
	6	800–1 000	$3\cdot10^{-5}$–$2\cdot10^{-2}$	0.86	>1 000	—	149
	6.4	927	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.9	—	[a]$3\cdot10^{-4}$/—	155
	6.8	880	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.67	—	—	11
	7.7	927	$3\cdot10^{-5}$–$2\cdot10^{-3}$	0.73	—	$7\cdot10^{-4}$/—	156
	6–12	927	$2\cdot10^{-5}$–$8\cdot10^{-3}$	0.94	—	10^{-4}/—, d = 8 μm	157
	5–16	815–927	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.7	600	$2\cdot10^{-4}$/927	158
	9	927	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.8	—	[a]10^{-4}/—	155
	11.5	927	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.75	—	[a]$8\cdot10^{-5}$/—	155
	20	927	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.7	—	[a]10^{-5}/—	155
Ti–6Al–4V–1Co–1Ni	3.4–22	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.82	550	$2\cdot10^{-4}$/815	158
Ti–6Al–4V–(X)Co, X = 0–2	6–12	750–950	10^{-5}–10^{-2}	1.0	—	[a]$3\cdot10^{-5}$/950, X = 1	159
Ti–6Al–4V–2Co	4–63	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.53	670	$2\cdot10^{-4}$/815	158
Ti–6Al–4V–1Fe–1Co	2.7–10	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.48	525	$2\cdot10^{-4}$/815	158
Ti–6Al–4V–1Fe–1Ni	3.4–20	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.68	550	$2\cdot10^{-4}$/815	158
Ti–6Al–4V–2Fe	3.5–8.6	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.54	650	$2\cdot10^{-4}$/815	158
Ti–6Al–4V–0.05H	2	750–870	10^{-5}–10^{-2}	0.61	610	10^{-4}/750	153
Ti–6Al–4V–0.07H	6.8	820	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.69	—	—	11
Ti–6Al–4V–0.1H	2	700–850	$5\cdot10^{-6}$–10^{-2}	0.6	545	10^{-3}/750	153
Ti–6Al–4V–0.14H	6.8	780	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.6	—	—	11
Ti–6Al–4V–0.17H	6.8	765	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.55	—	—	11

(*continued*)

Table 36.1 INVESTIGATIONS ON SUPERPLASTICITY OF METAL ALLOYS—*continued*

Material [wt %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Ti–6Al–4V–0.29H	6.8	735	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.28	—	—	11
Ti–6Al–4V–0.3H	—	700–750	10^{-5}–$9\cdot10^{-2}$	0.5	180	$5\cdot10^{-4}$/750	153
Ti–6Al–4V–0.40H	6.8	720	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.29	—	—	11
Ti–6Al–4V–0.53H	6.8	715	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.35	—	—	11
Ti–6Al–4V–(X)Ni, X = 0–2	6–12	750–950	10^{-5}–10^{-2}	0.98	—	[a] $3\cdot10^{-5}$/950, X = 1	159
Ti–6Al–4V–2Ni	3.5–35	815–871	$2\cdot10^{-5}$–$2\cdot10^{-2}$	0.85	720	$2\cdot10^{-4}$/815	158
Ti–6Al–4Zr–2Sn–2Mo	4.5	820–970	$2\cdot10^{-4}$–$2\cdot10^{-3}$	0.7	775	$2\cdot10^{-4}$/940	154
Ti–6.7Al–4.7Mo	0.06	500–600	10^{-6}–$3\cdot10^{-2}$	0.45	1 200	$2\cdot10^{-4}$/575	160
Ti–5.95Co–6.15Ni–5.05Al	<2	700–750	10^{-3}–10^{-1}	0.6	>1 500	$3\cdot10^{-3}$/700	161
Ti–10Co–4Al	0.5	550–750	$9\cdot10^{-6}$–$7\cdot10^{-2}$	0.48	>1 100	$2\cdot10^{-3}$/725	162
Ti–12.55Co–4.95Al	<2	700–750	10^{-3}–10^{-1}	0.6	>2 000	10^{-2}/700	161
Ti–11Mo–5.5Sn–4.2Zr	10	900	$3\cdot10^{-5}$–$3\cdot10^{-2}$	0.4	>300	$6\cdot10^{-5}$/—	163
Ti–10V–3Al–2Fe	<10	625–850	$4\cdot10^{-4}$–$4\cdot10^{-3}$	0.57	910	10^{-3}/700	164
Uranium Alloys							
U–1.82Mo	40	590–735	$3\cdot10^{-4}$–$5\cdot10^{-2}$	0.44	692	$3\cdot10^{-3}$/695	165
U–5.8Nb	1.2	600–750	$3\cdot10^{-4}$–$5\cdot10^{-3}$	0.85	658	$3\cdot10^{-4}$/685	166
Zinc Alloys							
Zn–1.1Al	1.1	20–220	10^{-5}–$4\cdot10^{-2}$	0.5	700	10^{-2}/102	27
Zn–18Al	0.6–2.2	250–350	$2\cdot10^{-6}$–$2\cdot10^{-2}$	0.53	—	—	167
Zn–20Al	0.5–1.5	20–250	$7\cdot10^{-5}$–$2\cdot10^{-1}$	0.55	>1 000	[a] $2\cdot10^{-1}$/250	168
Zn–22Al	1	60–200	$3\cdot10^{-7}$–$3\cdot10^{-1}$	0.5	—	—	169
	0.4–2	200–250	$3\cdot10^{-5}$–$2\cdot10^{-2}$	0.5	500	10^{-1}/250	170
	0.6–2.2	200–250	$2\cdot10^{-6}$–$2\cdot10^{-1}$	0.6	900	$2\cdot10^{-2}$/250	167
	2	220–300	$8\cdot10^{-4}$–$8\cdot10^{-1}$	0.44	1 200	—	171
	2.1	200	10^{-5}–10^{0}	0.41	—	—	172
	1.3–3.7	177–252	[c] 10^{-7}–10^{-1}	0.45	—	—	173
	2.4	200	$6\cdot10^{-6}$–$4\cdot10^{-1}$	—	1 800	—	174
	2.5	200	10^{-5}–10^{0}	—	2 900	10^{-2}/—	175
	2.6	230	10^{-5}–10^{-1}	0.5	2 550	$7\cdot10^{-2}$/—	176
	1–5	175–250	[c] $3\cdot10^{-9}$–$2\cdot10^{-2}$	0.5	—	[a] $2\cdot10^{-5}$–$2\cdot10^{-3}$	177
	2.3–4.6	136–230	10^{-7}–$2\cdot10^{-2}$	0.5	—	[a] 10^{-4}–10^{-2}	178
	2.5–4.2	150–230	10^{-5}–10^{0}	0.5	2 900	10^{-2}/230, d = 2.5 μm	179
	4–8	25–250	$3\cdot10^{-4}$–$3\cdot10^{-1}$	—	770	$3\cdot10^{-2}$/250	180
	—	25–350	$2\cdot10^{-5}$–10^{-2}	0.7	900	$2\cdot10^{-2}$/250	181
Zn–22.5Al	0.5–1.8	250	[b] $2\cdot10^{-6}$–2	0.5	—	—	182
Zn–23.55Al–0.43Cu	1	180	10^{-6}–10^{-1}	0.4	—	—	183
Zn–36Al	0.6–2.2	22–325	$2\cdot10^{-6}$–$2\cdot10^{-2}$	0.5	—	—	167
Zn–0.6Cu–0.1Ti	2.5	150–190	$8\cdot10^{-4}$	0.54	290	—/190	184
Zn–22Cu–20Al	2	20	$2\cdot10^{-4}$–$4\cdot10^{-2}$	0.34	180	—	185
Zn–0.2Mn	2–4	275	10^{-5}–1	0.17	102	—	186
Zn–0.7Mn	2–4	275	10^{-5}–1	0.39	620	—	186
Zn–1.2Mn	2–4	275	10^{-5}–1	0.36	500	—	186
Zn–1.5Mn	2–4	275	10^{-5}–1	0.27	132	—	186
Zn–2Mn	2–4	275	10^{-5}–1	0.22	150	—	186

[a] Optimum condition based on strain rate sensitivity, *m*, instead of elongation.
[b] Testing performed in compression.
[c] Testing performed in double-shear configuration.
[d] Dispersion-strengthened alloy.
[e] Tested under a 2 kV/cm applied electric field.

Table 36.2 INVESTIGATIONS ON INTERMETALLIC ALLOYS EXHIBITING SUPERPLASTICITY

Note that compositions are given in atomic %.

Material [at %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Iron Aluminide Alloys							
Fe_3Al Alloys							
Fe–27Al	100–800	600–800	10^{-4}–10^{-2}	0.33	180	10^{-4}/800	187
Fe–28Al–2Ti	100	800–900	$2{\cdot}10^{-4}$–$4{\cdot}10^{-3}$	0.42	330	10^{-3}/850	188
	100	850	10^{-4}–$2{\cdot}10^{-2}$	0.42	333	10^{-3}/—	189
FeAl Alloys							
Fe–36.5Al	>100	875–1 000	10^{-4}–$3{\cdot}10^{-1}$	0.33	—	—	190
Fe–36.5Al–1Ti	>100	875–1 000	10^{-4}–$3{\cdot}10^{-1}$	0.35	—	—	190
Fe–36.5Al–2Ti	350	900–1 000	10^{-4}–$3{\cdot}10^{-2}$	0.42	208	10^{-2}/1 000	191
	>100	875–1 000	10^{-4}–$3{\cdot}10^{-1}$	0.38	—	—	190
Iron Silicide (Fe_3Si) Alloys							
Fe–14Si–0.25B	72	800–950	$3{\cdot}10^{-5}$–$2{\cdot}10^{-3}$	0.33	200	$3{\cdot}10^{-5}$/800	192
Fe–18Si–0.25B	72	800–950	$3{\cdot}10^{-5}$–$2{\cdot}10^{-3}$	0.33	250	$3{\cdot}10^{-5}$/800	192
Nickel Aluminide Alloys							
NiAl Alloys							
NiAl	200	1 000–1 100	$2{\cdot}10^{-4}$–$2{\cdot}10^{-2}$	0.34	211	$2{\cdot}10^{-3}$/1100	193
Ni–28.5Al–20.4Fe	30–50	850–980	10^{-4}–10^{-2}	—	233	$5{\cdot}10^{-4}$/850	194
Ni_3Al Alloys							
Ni–15.5Al–7.4Cr–0.4Zr–0.09B	0.05	650–725	10^{-3}	—	375	10^{-3}/650	35
	6	1 000–1 100	10^{-4}–$2{\cdot}10^{-2}$	0.8	641	$9{\cdot}10^{-4}$/1100	195
	13–20	25–1100	$8{\cdot}10^{-4}$–$2{\cdot}10^{-2}$	0.6	638	$8{\cdot}10^{-4}$/1100	196
Ni–24Al–0.24B	1.6	700	$5{\cdot}10^{-5}$–$2{\cdot}10^{-3}$	0.45	155	$5{\cdot}10^{-5}$/—	197
Nickel Silicide (Ni_3Si) Alloys							
Ni–17.2Si–3.3V–1.1Mo	15	1 050–1 100	10^{-3}–$3{\cdot}10^{-2}$	0.5	500	10^{-3}/1 050	198
Ni–17.3Si–3.3V–2.3Mo	8–20	1 040–1 090	$9{\cdot}10^{-5}$–$2{\cdot}10^{-2}$	0.6	500	10^{-3}/1 070	199
	14	1 025–1 090	$6{\cdot}10^{-5}$–10^{-1}	0.6	665	$6{\cdot}10^{-4}$/1 090	200
Ni_3(Si,Ti)	4–14	800–900	$6{\cdot}10^{-5}$–10^{-3}	0.43	190	$6{\cdot}10^{-5}$/960	200
Titanium Aluminide Alloys							
TiAl Alloys							
Ti–40Al	0.5	900	$7{\cdot}10^{-4}$–$3{\cdot}10^{-3}$	0.56	—	—	201
Ti–43Al	5	1 000–1 100	10^{-5}–$2{\cdot}10^{-2}$	0.53	275	$3{\cdot}10^{-4}$/1 050	202
Ti–43.8Al–12.1V	<20	797–1147	$3{\cdot}10^{-4}$–10^{-1}	—	580	$3{\cdot}10^{-4}$/1147	203
Ti–45Al	0.5	900	$7{\cdot}10^{-4}$–$3{\cdot}10^{-3}$	0.52	—	—	201
Ti–45.5Al–2Cr–2Nb	3–12	900–1 200	10^{-3}	0.63	483	—/1 200	204
Ti–46Al	0.2–2.5	650–950	10^{-4}–10^{-1}	0.52	700	10^{-3}/900	205
Ti–46.1Al–3.1Cr	18	27–1 200	$5{\cdot}10^{-2}$–10^{-4}	0.58	383	$5{\cdot}10^{-4}$/1 200	175
Ti–47Al–2Cr–2Nb	0.5–5	650–1 000	10^{-6}–10^{-4}	0.5	310	$2{\cdot}10^{-5}$/800	206
Ti–47.3Al–1.9Nb–1.6Cr–0.5Si–0.4Mn	20	1 180–1 310	$2{\cdot}10^{-5}$–$2{\cdot}10^{-3}$	0.7	470	$8{\cdot}10^{-5}$/1280	207
Ti–48Al–2Cr–2Nb	10–22	1 100–1 300	[b] $2{\cdot}10^{-4}$–$2{\cdot}10^{-1}$	0.33	—	—	208
Ti–48.1Al–0.8Mo	2–5	25–1100	$2{\cdot}10^{-5}$–$2{\cdot}10^{-3}$	0.8	228	$4{\cdot}10^{-5}$/228	209
Ti–49.7Al	0.4	600–900	$2{\cdot}10^{-4}$–$2{\cdot}10^{-3}$	0.47	260	$8{\cdot}10^{-4}$/850	210
	0.4–22	20–850	$8{\cdot}10^{-4}$	—	260	—/850, d = 0.4 μm	211
	2–13	600–1 050	$2{\cdot}10^{-4}$–$8{\cdot}10^{-3}$	0.43	250	$8{\cdot}10^{-4}$/1 025	212
	2–15	900–1 050	10^{-4}–10^{-2}	0.43	250	$9{\cdot}10^{-4}$/1 025	213
Ti–50Al	8	1 025	10^{-4}–10^{-2}	0.43	250	10^{-3}/1 025	163
	0.3	900	$7{\cdot}10^{-4}$–$3{\cdot}10^{-3}$	0.37	—	—	201

(*continued*)

Table 36.2 INVESTIGATIONS ON INTERMETALLIC ALLOYS EXHIBITING SUPERPLASTICITY—*continued*

Material [at %]	*Grain size* [μm]	*Temp. range* [°C]	*Strain rate range* [s^{-1}]	*m*	e_f [%]	*Optimum condition* [s^{-1}]/[°C]	*Ref.*
Ti_3Al Alloys							
Ti–24Al–11Nb	3	920–1 040	$2{\cdot}10^{-5}$–$5{\cdot}10^{-3}$	0.7	810	$5{\cdot}10^{-5}$/980	214
	4	960–1 020	$3{\cdot}10^{-6}$–10^{-2}	0.6	520	$3{\cdot}10^{-4}$/980	215
Ti–24.4–10.6Nb–3V–1Mo	3.4	850–1100	$5{\cdot}10^{-5}$–$5{\cdot}10^{-3}$	0.7	—	—	216
Ti–25Al–10Nb–3V–1Mo	1–10	950–1 040	$8{\cdot}10^{-6}$–$7{\cdot}10^{-2}$	0.55	800	$8{\cdot}10^{-5}$/950	217
	2–12	900–1 000	$2{\cdot}10^{-5}$–10^{-2}	0.75	1 500	$2{\cdot}10^{-4}$/960	218
	<20	950–1 010	$2{\cdot}10^{-5}$–$5{\cdot}10^{-3}$	0.62	570	$2{\cdot}10^{-4}$/980	219
Ti–30Al	1.5	900	$7{\cdot}10^{-4}$–$3{\cdot}10^{-3}$	0.41	—	—	201

[b] Testing performed in compression.

Table 36.3 SUMMARY OF RESEARCH ON SUPERPLASTIC METAL MATRIX COMPOSITES

Grain size, d, refers to the matrix material, and V_f denotes the volume fraction of the reinforcing phase.

Matrix	*d* [μm]	*Reinforcement*	V_f [%]	*m*	e_f [%]	*Optimum condition* T [°C]	*Optimum condition* $\dot{\varepsilon}$ [s^{-1}]	*Ref.*
Al	2	AlN_P, 1.8 μm	—	0.3	200	640[f]	10^{-1}	220
Al–2014	2.5	Al_2O_{3P}	20	0.45	210	500[f]	$2{\cdot}10^{-2}$	221
	—	SiC_P, 6–10 μm	15	0.4	442	480	10^{-4}	222
Al–2024	7	SiC_P, 10 μm	10	0.44	685	515	$5{\cdot}10^{-4}$	223
	3–13	SiC_W	20	0.5	150	550	10^{0}	224
Al–2124	<10	SiC_P, 3 μm	17.8	0.41	425	490	$8{\cdot}10^{-2}$	225
	—	SiC_W	20	0.33	300	525[f]	$3{\cdot}10^{-1}$	226
	2	β- Si_3N_{4W}, 0.5–1.5 by 10–20 μm	20	0.5	225	525[f]	$2{\cdot}10^{-1}$	227
	—	Si_3N_{4W}, 0.1–15 by 10–30 μm	20	>0.3	280	545	$4{\cdot}10^{-2}$	228
	—	Si_3N_{4P}, <1 μm	20	>0.3	830	515	$4{\cdot}10^{-2}$	228
Al–2324	1	SiC_W	20	0.55	520	520	$5{\cdot}10^{-2}$	229
Al–5052	—	Si_3N_{4P}, <0.2 μm	20	>0.3	700	545	10^{0}	228
Al–6061	—	AlN_P, 1–10 μm	~20	0.5	509	600[f]	$9{\cdot}10^{-1}$	230
	—	SiC_W	20	0.32	300	550	10^{-1}	231
	1.9	Si_3N_{4P}, 0.5 μm	20	0.5	500	560[f]	$5{\cdot}10^{-4}$	232
	<3	Si_3N_{4P}, 1 μm	20	0.45	450	545[f]	10^{-1}	233
	—	β-Si_3N_{4P}	25	0.46	600	545[f]	$2{\cdot}10^{-1}$	234
	—	Si_3N_{4W}, 0.1–15 by 10–30 μm	20	>0.3	285	560	10^{-1}	228
	—	Si_3N_{4P}, <0.2 μm	20	>0.3	620	560	$2{\cdot}10^{0}$	228
	—	Si_3N_{4P}, <1 μm	20	>0.3	615	545	$2{\cdot}10^{-1}$	228
Al–7064	—	α-Si_3N_{4W}, 0.1–15 by 5–200 μm	20	0.42	150	525	$3{\cdot}10^{-1}$	235
	—	β-Si_3N_{4W}, 0.5–15 by 10–20 μm	15	0.22	150	545	$2{\cdot}10^{-1}$	235
	—	β-Si_3N_{4W}, 0.5–15 by 10–20 μm	20	0.35	150	525	$2{\cdot}10^{-1}$	235
	—	β-Si_3N_{4W}, 0.5–15 by 10–20 μm	25	0.24	70	605	10^{0}	235
	—	Si_3N_{4W}, 0.1–15 by 10–30 μm	20	>0.3	380	560	10^{-1}	228
	—	Si_3N_{4P}, <1 μm	20	>0.3	330	545	10^{0}	228

(*continued*)

Table 36.3 SUMMARY OF RESEARCH ON SUPERPLASTIC METAL MATRIX COMPOSITES—*continued*

Matrix	*d* [μm]	*Reinforcement*	V_f [%]	*m*	e_f [%]	*Optimum condition* T [°C]	*Optimum condition* $\dot{\varepsilon}$ [s^{-1}]	*Ref.*
Al–7475	—	SiC_P, 0.6 μm	20	0.33	200	580[f]	10^{-1}	236
	7–22	SiC_P, 6–10 μm	15	0.43	395	480	$8{\cdot}10^{-4}$	222
	5–10	SiC_P, 10 μm	16	0.67	315	515	$2{\cdot}10^{-4}$	237
Al–IN9021	0.45	SiC_P, 2 μm	15	0.5	610	550[f]	$5{\cdot}10^{0}$	80
Mg–ZK60	1.8	SiC_P, 0.5 μm	17	0.5	450	350	10^{-1}	238
	2	SiC_P, 2 μm	17	0.5	450	350	10^{-1}	239
	0.3	SiC_P, 2 μm	17	0.33	360	450	10^{0}	240
	0.5	SiC_P, 2 μm	17	>0.5	320	400	10^{0}	241
Mg–5Zn	—	TiC_P, 2–5 μm	10	0.33	200	470	$2{\cdot}10^{-2}$	242
	—	TiC_P, 2–5 μm	20	0.43	300	470	$7{\cdot}10^{-2}$	242
Mg–10.6Si–4Al	1.4	Mg_2Si_P, 1 μm	20	0.3	346	500[f]	10^{0}	243
Ti–6Al–4V	<5	TiC_P, 5 μm	10	0.64	270	925[f]	$2{\cdot}10^{-4}$	244
	<5	TiC_P, 5 μm	20	0.56	270	925[f]	$2{\cdot}10^{-4}$	244
	<5	TiN_P, 5 μm	10	0.57	270	925[f]	$2{\cdot}10^{-4}$	244
	<5	TiN_P, 5 μm	20	0.62	270	925[f]	$2{\cdot}10^{-4}$	244

[f] only one temperature was investigated.

Table 36.4 SUMMARY OF RESEARCH ON INTERNAL STRESS SUPERPLASTICITY

The source of internal mismatch stress is denoted by C = thermal expansion mismatch between coexisting phases in a composite, A = thermal expansion mismatch between adjacent grains in an anisotropic material, and T = phase transformation volume mismatch.

Material	*Mismatch*	*Temp. range* [°C]	*Frequency* [hr^{-1}]	e_f [%]	*Ref.*
Al–0.8Be	C	300–400	36–180	200	245
Al–0.8Be	C	250–350	90	—	246
		300–400			
		350–450			
		400–500			
Al–12Si	C	165–300	15	220	247
		125–235			
		141–281			
		164–334			
Al–Al_3Ni	C	400–500	90	>120	248
Al/5 vol% SiC_P	C	130–450	8.5	—	249
Al/10 vol% SiC_P		130–350			
Al/20 vol% SiC_P		80–200			
Al/30 vol% SiC_P					
Al/40 vol% SiC_P					
Al/20 vol% SiC_P	C	40–150	8.5	>90	250
		110–350			
Al/20 vol% SiC_P	C	80–200	8.5	>150	251
Al/30 vol% SiC_P		130–350			
		130–450			
Al/20 vol% SiC_P	C	100–300	6	—	252
		115–350			
Al–2024/10 vol% SiC_W	C	100–450	18	—	253
Al–2024/20 vol% SiC_W					
Al–2024/20 vol% SiC_W	C	100–450	18	300	254
Al–2024/20 vol% SiC_W	C	100–450	20–22.5	300	255
Al–2124/20 vol% SiC_W	C	100–350	7	220	256
Al–6061/20 vol% SiC_W	Compressibility mismatch during pressure cycling at 420°C			96	257

(*continued*)

Table 36.4 SUMMARY OF RESEARCH ON INTERNAL STRESS SUPERPLASTICITY—*continued*

Material	*Mismatch*	*Temp. range* [°C]	*Frequency* [hr^{-1}]	e_f [%]	*Ref.*
Al–6061/20 vol% SiC_W	C	100–450	18	1 300	254
Al–6061/20 vol% SiC_W	C	100–450	18	1 400	258
Al–7090/25 vol% SiC_P	C	135–200 100–200 70–200	6	—	259
Co	T	367–467	—	—	260
Co	T	Single heating excursions, 1–4 K/min		—	261
Cu–Zn Alloys	T	−180–20	—	—	262
Fe	T	750–850 800–1 000	—	—	263
Fe	T	860–930	1–6	—	264
Fe	T	770–960	50	—	265
Fe	T	860–960	—	—	260
Fe	T	700–900	6–10	>454	266
Fe/10 vol% TiC_P	T	Various, 700–950	6–30	231	266
Fe/20 vol% TiC_P	T	700–900	6–10	—	266
Fe–0.2C	T	600–900	36	—	267
Fe–0.2C	T	740–800	72	—	264
Fe–0.39C	T	673–850	—	—	260
Fe–AISI 1018	T	540–815	50	510	268
Fe–AISI 1045	T	540–815	50	500	268
Fe–AISI 1095	T	540–815	50	580	268
Fe–AISI 52100	T	540–815	50	695	268
Fe–0.91C–0.45Mn–0.12Si	T	538–815	~20	>490	102
Fe–2 at% Cr	T	Single cooling excursions, 1–65 K/s		—	269
Mg–AZ31	A	275–375 300–400 325–425	90	—	270
Ni–Mar–M247	C	1 100–1 200	180	>100	271
NiAl–Cr	C	950–1 150 975–1 175 1 000–1 200	90	—	272
NiAl/10–20 vol% ZrO_2	T	700–1 200 850–1 250 750–1 150	7.5–15	>30	273
Ti	T	Single excursions, 110–420 K/h		—	274
Ti	T	800–900	0.18–90	—	275, 276
Ti	T	650–925	120	—	277
Ti	T	832–932	—	—	260
Ti	T	830–1030	4.2	—	278
Ti	T	Cycles of chemical composition through cyclic hydrogen introduction		—	279
Ti/10 vol% TiC	T	830–1 010	4.2	135	278, 280
Ti–6Al–4V	T	760–981	120	—	277
Ti–6Al–4V	T	840–1 030	7.5	260–398	281, 282
Ti–6Al–4V	T	Various, 840–990	7.5	—	283
Ti–6Al–4V/5 vol% TiB_W	T	840–1 030	7.5	390	284
Ti–6Al–4V/10 vol% TiB_W	T	840–1 030	7.5	260	282
Ti_3Al, Super α_2	T	950–1 150	4–15	610	285

(*continued*)

Table 36.4 SUMMARY OF RESEARCH ON INTERNAL STRESS SUPERPLASTICITY—*continued*

Material	*Mismatch*	*Temp. range* [°C]	*Frequency* [hr^{-1}]	e_f [%]	*Ref.*
U	T	Various, 200–450	50–170	—	286
U	T	613–713 720–820	—	—	260
Zn	A	60–150 110–330	14.4 8.5	—	251
Zn	A	50–250 100–300 150–350	12–180	>200	287
Zn	A	200–300 230–330 250–350 270–370 290–390	90	—	288
Zn–30 vol% Al_2O_{3P}	A,C	100–300 150–350	120	—	289
Zr	T	750–950	—	—	263
Zr	T	813–913	—	—	260
Zr	T	810–910 810–940	6–15	>270	290
Zr/2 vol% ZrH_2	T	23–425	0.036	—	291

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